

Natural Logarithmic function

(1)

Recall: The log. function is the inverse of the exponential function.

ie: $y = e^x \Rightarrow e^y = x$?

(OR) $y = \log_e x$ (base "e")

log with base "e" is given a "special" name: $y = \log_e x = \ln(x)$

↑
read "lawn of x"

Given $y = \ln(x)$ find y' .

* Rewrite $y = \ln(x)$ to be $e^y = x$
 (log form)
 (exp form)

* Let $y = g(x)$:

$$e^{g(x)} = x$$

* Differentiate:

$$e^{g(x)} \cdot g'(x) = 1$$

$$g'(x) = \frac{1}{e^{g(x)}} = \frac{1}{e^y} = \frac{1}{x}$$

$y = \ln(x)$

$$\therefore y' = \frac{1}{x}, x > 0$$

In general, if $f(x) = \ln(\underline{g(x)})$, then

$$f'(x) = \frac{g'(x)}{g(x)}$$

$g(x) \Rightarrow$ polynomial
 $\Rightarrow$ root
 $\Rightarrow$ trig.
 $\rightarrow$ exp
 $\rightarrow$ rational

notes:

* (1) $\ln e = 1 \rightarrow e^1 = e$ *

* (2) $\ln 1 = 0 \rightarrow \underline{e^0 = 1}$ *

why? $\log_e e = 1$

$\log_e 1 = 0$

$\log_a a = 1$

Example 1: Differentiate.

a) $y = \ln(4x)$

$y' = \frac{4}{4x}$

$y' = \frac{1}{x}$

$\ln(g(x)) \rightarrow \frac{g'(x)}{g(x)}$

$\ln(nx) = \frac{1}{x} \quad n \neq 0$

b) $y = \ln(x^2 + 5x - 7)$ log function

$y' = \frac{2x + 5}{x^2 + 5x - 7}$ rational function

Not in domain $\ln(0)$ $\ln(0)$ $\ln(0)$

c) $y = [\ln(x)]^5$

$y' = 5 [\ln(x)]^4 \cdot \frac{1}{x}$

$\ln(x^5)$

$\rightarrow 5 \ln(x)$ law of log.

chain rule.

$y' = \frac{5 [\ln(x)]^4}{x}$

crit # : $x = 1$

$x = 0$ $f'(0)$ DNE not a critical #

d) $y = 2x \cdot \ln(x)$ product rule

$y' = (2x)' \ln(x) + 2x \cdot \frac{1}{x}$

$y' = 2 \ln(x) + 2$

$y' = 2 (\ln(x) + 1)$ crit #:

$2(\ln(x) + 1) = 0$

$\ln(x) + 1 = 0$

$\ln(x) = -1$

$e^{-1} = x \Rightarrow x = \frac{1}{e}$

$$x = -1 \rightarrow \ln \sqrt{-\frac{2}{0}}$$

UND.

$$x = 1 \rightarrow \ln \sqrt{0}$$

UND.

4

$$e) y = \ln \left[\sqrt{\frac{x-1}{x+1}} \right]$$

$$y' = \frac{\left(\sqrt{\frac{x-1}{x+1}} \right)'}{\sqrt{\frac{x-1}{x+1}}}$$

do not
do this!
way.

$$y = \ln \sqrt{x-1} - \ln \sqrt{x+1}$$

$$y = \ln (x-1)^{\frac{1}{2}} - \ln (x+1)^{\frac{1}{2}}$$

$$\ln \left(\frac{A}{B} \right)$$

$$\log A - \log B$$

$$y = \frac{1}{2} \ln(x-1) - \frac{1}{2} \ln(x+1)$$

$$y' = \frac{1}{2(x-1)} - \frac{1}{2(x+1)}$$

$$\frac{1}{x+1}$$

$$y' = \frac{x+1 - (x-1)}{2(x^2-1)}$$

$$y' = \frac{2}{2(x^2-1)}$$

$$y' = \frac{1}{x^2-1}$$

critical
#s = NONE!

$x = \pm 1 \in$ not in domain

$$f) y = \ln(\sin(x))$$

$$g) y = e^{-x^2} \cdot \sin(\ln x)$$

(5)

$$y' = \frac{\cos(x)}{\sin(x)}$$

$$y' = -2xe^{-x^2} \sin(\ln x) + \frac{e^{-x^2} \cdot \cos(\ln x)}{x} \quad (3)$$

$$y' = \cot(x)$$

Ex 2: Determine the equation of the normal line to $y = \frac{\ln(x^2)}{3x}$ at $x=1$.

$$y' = \frac{\left(\frac{2x}{x^2}\right)(3x) - \ln(x^2) \cdot 3}{(3x)^2}$$

$$\frac{\ln(1)}{3} = \frac{0}{3} = 0$$

$\swarrow$ $6 \ln x$

$$y' = \frac{6 - 3 \ln(x^2)}{9x^2}$$

$\swarrow$ $6 \ln x$ (OR) $\ln(x^6)$

$\swarrow$ $\ln 1 = 0$

$$m_t|_{x=1} = \frac{6-0}{9} = \frac{2}{3} \Rightarrow m_{\perp} = -\frac{3}{2}$$

$\swarrow$ neg. reciprocal of

$$y - y_1 = m_{\perp}(x - x_1)$$

$$y - 0 = -\frac{3}{2}(x - 1)$$

$$y = -\frac{3}{2}(x - 1)$$